

Basic Linear Algebra

with Applications to Data Analysis

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Refresher I

— Basic Notions —

- ▶ Groups, Fields, Vector Spaces
- ▶ Subspaces, Spanning Sets, Bases
- ▶ Direct Sums, Linear Complements
- ▶ Linear Functions, Isomorphisms
- ▶ Rank, Kernel, Rank-Nullity Theorem

Groups, Fields & Vector Spaces

Definition (Group)

A **group** is a set G with a binary operation $\cdot: G \times G \rightarrow G$ s.t.

- ▶ **Associativity:** $\forall a, b, c \in G : (a \cdot b) \cdot c = a \cdot (b \cdot c)$
- ▶ **Identity element:** $\exists! e \in G \forall a \in G : a \cdot e = a = e \cdot a$
- ▶ **Inverse element:** $\forall a \in G \exists! a^{-1} \in G : a \cdot a^{-1} = e = a^{-1} \cdot a$

The group is **abelian/commutative** if

$$a + b = b + a \quad \forall a, b \in G.$$

Groups, Fields & Vector Spaces

Definition (Field)

A **field** is a set $\mathbb{K}$ with binary operations $+, \cdot: \mathbb{K} \times \mathbb{K} \rightarrow \mathbb{K}$ s.t.

- ▶ $(\mathbb{K}, +)$ is an abelian group with identity element 0;
- ▶ $\mathbb{K}_* := \mathbb{K} \setminus \{0\}$ is an abelian group with identity element 1;
- ▶ $\forall x, y, z \in \mathbb{K}$:

$$x \cdot (y + z) = x \cdot y + x \cdot z.$$

Groups, Fields & Vector Spaces

Definition (Vector Space)

A **vector space** over a field $\mathbb{K}$ is a set V with a binary operation $+: V \times V \rightarrow V$ and a *scalar multiplication* $\cdot: \mathbb{K} \times V \rightarrow V$ s.t.

- ▶ $(V, +)$ is an abelian group;
- ▶ $+$ and $\cdot$ are compatible in the sense that for all $\lambda, \lambda_1, \lambda_2 \in \mathbb{K}$ and $v, w \in V$ holds
 - (i) $1 \cdot v = v$;
 - (ii) $(\lambda_1 + \lambda_2) \cdot v = (\lambda_1 \cdot v) + (\lambda_2 \cdot v)$.
 - (iii) $(\lambda_1 \lambda_2) \cdot v = \lambda_1 \cdot (\lambda_2 \cdot v)$;
 - (iv) $\lambda \cdot (v_1 + v_2) = (\lambda \cdot v_1) + (\lambda \cdot v_2)$;

Subspaces & Spanning Sets

Definition

A **linear subspace** of a vector space V over $\mathbb{K}$ is a subset $U \subset V$ that is closed under addition and scalar multiplication:

- ▶ $0 \in U$;
- ▶ if $u_1, u_2 \in U$ then $u_1 + u_2 \in U$;
- ▶ if $u \in U$ then $\lambda \cdot u \in U$ for all $\lambda \in \mathbb{K}$.

Definition

The **span** of a set $S \subset V$ is

$$\text{span}(S) := \left\{ \sum_{i=1}^N \lambda_i \cdot s_i \mid N \in \mathbb{N}, s_i \in S, \lambda_i \in \mathbb{K} \right\}.$$

If $\text{span}(S) = V$, then S is a **spanning set** of V .

Bases of Vector Spaces

Definition

A set $\mathcal{B} \subset V \setminus \{0\}$ is called **linear independent** if for all $N \in \mathbb{N}$ and $b_1, \dots, b_N \in \mathcal{B}$ holds:

$$\lambda_1 b_1 + \dots + \lambda_N b_N = 0 \quad \Longleftrightarrow \quad \lambda_1 = \dots = \lambda_N = 0.$$

If $\mathcal{B}$ is also a *spanning set* of V , then $\mathcal{B}$ is called a **basis** of V . The **dimension** of V is the cardinality $\dim_{\mathbb{K}}(V) = |\mathcal{B}|$. In particular, V is **finite dimensional** if $\dim_{\mathbb{K}}(V) < \infty$.

New from Old: Union ($\cup$) & Intersection ($\cap$)

Proposition

Let V_1, V_2 be subspaces of V . Then

- ▶ $V_1 \cap V_2$ is *always* a subspace of V ;
- ▶ $V_1 \cup V_2$ is a linear subspace of V *if and only if* $V_1 \subset V_2$ or $V_2 \subset V_1$.

Proposition

Let $\mathcal{S} \subset V$. The span of $\mathcal{S}$ is the smallest linear subspace of V containing $\mathcal{S}$ in the sense that

$$\text{span}(\mathcal{S}) = \bigcap_{\substack{\mathcal{S} \subset U \subset V \\ \text{subspace}}} U.$$

New from Old: Sum (+) & Direct Sum ($\oplus$)

Definition (Direct Sum)

Let V, W be vector spaces over $\mathbb{K}$. The **direct sum** $V \oplus W$ is the set $V \times W$ endowed with the

- ▶ *addition*: $(v_1, w_1) + (v_2, w_2) = (v_1 + v_2, w_1 + w_2)$;
- ▶ *scalar mult.*: $\lambda \cdot (u, w) = (\lambda v, \lambda w)$.

Definition (Sum & Linear Complement)

Let V_1, V_2 be linear subspaces of V .

- ▶ The **sum** of V_1 and V_2 is the subspace $V_1 + V_2 := \text{span}(V_1 \cup V_2)$.
- ▶ V_1 and V_2 are **linear complements (in V)** if $V_1 \oplus V_2$ is isomorphic to V .

Linear Functions

Definition

Let V, W be vector spaces over $\mathbb{K}$.

- ▶ A **linear function** is a function $f: V \rightarrow W$ such that

$$f(\lambda_1 v_1 + \lambda_2 v_2) = \lambda_1 f(v_1) + \lambda_2 f(v_2)$$

for all $\lambda_1, \lambda_2 \in \mathbb{K}$ and $v_1, v_2 \in V$.

- ▶ A linear function is called **isomorphism** if it admits a linear inverse $f^{-1}: W \rightarrow V$, i.e., f^{-1} is linear and $f^{-1} \circ f = \text{id}_V$.
In this case we call V and W *(linear) isomorphic* and write $V \simeq W$.

Proposition

Let $\mathcal{B} \subset V$ be a basis. Every linear map $f: V \rightarrow W$ is determined by its action on $\mathcal{B}$, i.e., if $w_b \in W$ for $b \in \mathcal{B}$, then

$$f(b) := w_b$$

defines f completely.

Proposition

The set $\text{Lin}(V, W)$ of linear functions from V to W is a vector space. If $\dim(V), \dim(W) < \infty$, then $\dim(\text{Lin}(V, W)) = \dim(V) \cdot \dim(W)$.

Proposition

The set $\text{GL}(V)$ of isomorphisms $V \rightarrow V$ is a group.

Basics of Matrices

Definition

An (m, n) -matrix over a field $\mathbb{K}$ is a map $A: \{1, \dots, m\} \times \{1, \dots, n\} \rightarrow \mathbb{K}$.
We usually write

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} = (a_{ij})_{mn} = (a_{ij})$$

For $i \in \{1, \dots, m\}$ and $j \in \{1, \dots, n\}$ is $(a_{i1} \dots a_{in})$ the i th row and $\begin{pmatrix} a_{1j} \\ \vdots \\ a_{mj} \end{pmatrix}$ the j th column of A .

The set of all (m, n) -matrices of $\mathbb{K}$ is denoted by $\mathbb{K}^{m,n}$.

Basics of Matrices

Definition

- ▶ The **matrix product** $\cdot: \mathbb{K}^{m,n} \times \mathbb{K}^{n,p} \rightarrow \mathbb{K}^{m,p}$ is

$$(a_{ij})_{mn} \cdot (b_{ij})_{np} := \left(\sum_{k=1}^n a_{ik} b_{kj} \right)_{mp}$$

- ▶ The *transpose* of a matrix $A = (a_{ij})_{mn} \in \mathbb{K}^{m,n}$ is $A^T = (\tilde{a}_{ij})_{nm} \in \mathbb{K}^{n,m}$ with $\tilde{a}_{ij} = a_{ji}$.

Matrices & Linear Functions

Proposition

- ▶ Each $A \in \mathbb{K}^{m,n}$ defines a linear function $L_A: \mathbb{K}^n \rightarrow \mathbb{K}^m$ via

$$L_A(x) := A \cdot v, \quad v \in \mathbb{K}^n.$$

- ▶ For all $A \in \mathbb{K}^{m,n}$ and $B \in \mathbb{K}^{n,p}$ holds $L_A \circ L_B = L_{A \cdot B}$.

Matrices & Linear Functions

Proposition

Let $\mathcal{B} \subset V$, $\mathcal{B}' \subset W$ be bases with $\dim V = n$ and $\dim W = m$.

Then for $f \in \text{Lin}(V, W)$ and all $j \in \{1, \dots, n\}$ there are $a_{1j}, \dots, a_{mj} \in \mathbb{K}$ such that

$$f(b_j) = a_{1j}b'_1 + \dots + a_{mj}b'_m.$$

The **matrix representation of f** is the matrix $[f]_{\mathcal{B}, \mathcal{B}'} := (a_{ij})_{ij}$.

It satisfies that for $v = \lambda_1 b_1 + \dots + \lambda_n b_n \in \mathbb{K}^n$ and

$\mu := L_{[f]_{\mathcal{B}, \mathcal{B}'}}(\lambda_1, \dots, \lambda_n) \in \mathbb{K}^m$ holds

$$f(v) = \mu_1 b'_1 + \dots + \mu_m b'_m.$$

Rank-Nullity of Linear Functions

Definition

Let $f: V \rightarrow W$ be a linear function and $A \in \mathbb{K}^{m,n}$.

- ▶ The **kernel** of f is

$$\ker(f) := \{v \in V \mid f(v) = 0\}.$$

If $\dim(V) < \infty$, then the **nullity** of f is $\dim \ker(f)$.

- ▶ The **image** of f is

$$\operatorname{im}(f) := \{f(v) \in W \mid v \in V\}.$$

If $\dim(V) < \infty$, then the **rank** of f is $\dim \operatorname{im}(f)$.

- ▶ We define $\ker(A) := \ker(L_A)$ and $\operatorname{im}(A) := \operatorname{im}(L_A)$.

Rank-Nullity of Linear Functions

Theorem (Rank-Nullity)

Let $f: V \rightarrow W$ be a linear function and $A \in \mathbb{K}^{m,n}$. Then

- ▶ $V \simeq \ker(f) \oplus \operatorname{im}(f)$
- ▶ If $\dim(V) < \infty$, then $\dim(V) = \dim \ker(f) + \dim \operatorname{im}(f)$
- ▶ $n = \dim \ker(L_A) + \dim \operatorname{im}(L_A)$

The Determinant

Definition

The **determinant** $\det: \mathbb{K}^{n,n} \rightarrow \mathbb{K}$ is

$$\det A := \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{1\sigma(1)} \cdots a_{n\sigma(n)}.$$

Proposition

Some of the properties of the determinant are:

- ▶ $\forall A, B \in \mathbb{K}^{n,n}$: $\det(A \cdot B) = \det(A) \det(B)$
- ▶ $\det(A) = 0 \Leftrightarrow A \in \operatorname{GL}(\mathbb{K}; n)$
- ▶ $\det$ is multi-linear, but not linear!

The Determinant: Adjugate Formula

Definition

- ▶ Let $A \in \mathbb{K}^{n,n}$. The *ij-minor matrix* $A^{ij} \in \mathbb{K}^{n-1,n-1}$ is obtained by deleting the i th row and j th column of A .
- ▶ The *adjugate* $\text{adj}(A) \in \mathbb{K}^{n,n}$ of $A \in \mathbb{K}^{n,n}$ is given by

$$\text{adj}(A)_{ij} := (-1)^{i+j} \det(A^{ji}).$$

Proposition

For all $A \in \mathbb{K}^{n,n}$ holds

$$A \text{adj}(A) = \text{adj}(A)A = \det(A)I_n.$$

In particular, for $A \in \text{GL}(\mathbb{K}; n)$ we have

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A).$$

The Determinant: Isomorphisms

Proposition

A map $f \in \text{Lin}(V, W)$ between finite dimensional vector spaces is an isomorphism *if and only if* there exists a representation as a square matrix $[f]_{\mathcal{B}_V, \mathcal{B}_W} \in \mathbb{K}^{n,n}$ with $\det([f]_{\mathcal{B}_V, \mathcal{B}_W}) \neq 0$.

Moreover, the value of $\det([f]_{\mathcal{B}_V, \mathcal{B}_W})$ is independent of the choice of bases $\mathcal{B}_V$ and $\mathcal{B}_W$.

Refresher II

— Eigenvalue Problems —

- ▶ Eigenvalues and Eigenvectors
- ▶ Characteristic Polynomial
- ▶ Criteria for Diagonalizability
- ▶ Jordan Decomposition

Eigenvalues & Eigenvectors

Definition

- ▶ An **eigenvalue-eigenvector pair** of $f \in \text{Lin}(V, V)$ is are $\lambda \in \mathbb{K}$ and $v \in V \setminus \{0\}$ s.t.

$$f(v) = \lambda v.$$

- ▶ The eigenpairs of $A \in \mathbb{K}^{n,n}$ are the eigenpairs of $L_A \in \text{Lin}(\mathbb{K}^n, \mathbb{K}^n)$.

Proposition

The **eigenspace** $\text{Eig}(f; \lambda) \subset V$ of f for $\lambda \in \mathbb{K}$ is the vector space of eigenvectors of f for λ . In particular,

$$\text{Eig}(f; \lambda) = \ker(\lambda \text{id}_V - f).$$

Characteristic Polynomial

Definition

Let $A \in \mathbb{K}^{n,n}$. The **characteristic polynomial** of A is

$$\chi_A(t) := \det(tI_n - A) \in \mathbb{K}[t].$$

Proposition

- ▶ $\chi_A(t)$ is monic of degree n . In particular,

$$\chi_A(t) = t^n - \operatorname{tr}(A) + \cdots + (-1)^n \det(A).$$

- ▶ $\lambda \in \mathbb{K}$ is an eigenvalue of A *if and only if* λ is a root of $\chi_A(t)$.

Criteria for Diagonalizability

Definition

A square matrix $A \in \mathbb{K}^{n,n}$ is called **diagonalizable** if there exists $S \in \text{GL}(\mathbb{K}; n)$ and $\lambda_1, \dots, \lambda_n \in \mathbb{K}$ such that

$$S^{-1}AS = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}.$$

Proposition

Let $A \in \mathbb{K}^{n,n}$.

- ▶ A is diagonalizable *if and only if* $\mathbb{K}^n$ possesses a basis of eigenvectors of A .
- ▶ If $\chi_A(t)$ has n distinct roots, then A is diagonalizable.

Minimal Polynomial

Definition

Let $A \in \mathbb{K}^{n,n}$. The **minimal polynomial** $\mu_A(t) \in \mathbb{K}[t]$ of A is the monic polynomial of smallest degree that satisfies $\mu_A(A) = 0$.

Proposition

Let $A \in \mathbb{K}^{n,n}$.

- ▶ The characteristic polynomial $\chi_A(t)$ is a multiple of $\mu_A(t)$, i.e., $\exists q(t) \in \mathbb{K}[t]$ with $\chi_A(t) = q(t)\mu_A(t)$.
- ▶ A is diagonalizable *if and only if* $\mu_A(t)$ splits over $\mathbb{K}$ and has distinct roots, i.e., $\exists m < n, \lambda_i \in \mathbb{K}$ with $\lambda_i \neq \lambda_j$ s.t. $\mu_A(t) = \prod_{i=1}^m (t - \lambda_i)$.

Corollary

Every **projector** $P \in \text{Lin}(V, V)$, $\dim(V) < \infty$, is diagonalizable with eigenvalues in $\{0, 1\}$.

Jordan Decomposition

Definition

The $n \times n$ Jordan block for $\lambda \in \mathbb{K}$ is $J_{\lambda,n} := \begin{pmatrix} \lambda & 1 & & \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ & & & \lambda \end{pmatrix} \in \mathbb{K}^{n,n}$.

Theorem (Jordan Decomposition)

Let $A \in \mathbb{K}^{n,n}$ and assume that $\chi_A(t)$ splits over $\mathbb{K}$. Then there is a $k \in \mathbb{N}$, $(\lambda_i, n_i) \in \mathbb{C} \times \mathbb{N}$ and $S \in \text{GL}(\mathbb{K}; n)$ s.t.

$$S^{-1}AS = \begin{pmatrix} J_{\lambda_1, n_1} & & 0 \\ & \ddots & \\ 0 & & J_{\lambda_k, n_k} \end{pmatrix}.$$

Jordan Decomposition: Some Corollaries

Theorem (Fundamental Theorem of Algebra)

Every polynomial with coefficients in $\mathbb{C}$ splits, i.e., for every $p(t) \in \mathbb{C}[t]$ there exist $c, \lambda_1, \dots, \lambda_n \in \mathbb{C}$ with $p(t) = c(t - \lambda_1) \cdot \dots \cdot (t - \lambda_n)$.

Corollaries

- ▶ Every $A \in \mathbb{C}^{n,n}$ possesses a Jordan decomposition.
- ▶ For $A \in \mathbb{C}^{n,n}$ holds $\det(A) = \prod_{i=1}^k \lambda_i^{n_i}$ and $\operatorname{tr}(A) = \sum_{i=1}^k n_i \lambda_i$.

Jordan Decomposition: Computation

Let $A \in \mathbb{K}^{n,n}$.

- (i) Factorize $\chi_A(t)$: find $m, n_i \in \mathbb{N}$ and $\lambda_i \in \mathbb{K}$ s.t. $\chi_A(t) = \prod_{i=1}^m (t - \lambda_i)^{n_i}$.
- (ii) $\forall i$: find $k_i \in \mathbb{N}$ s.t. $(I_n - \lambda_i A)^{k_i-1} \neq 0$ and $(I_n - \lambda_i A)^{k_i} = 0$.
- (iii) $\forall i$: Compute a linear complement V_i of

$$\ker(I_n - \lambda_i A)^{k_i-1} \subset \ker(I_n - \lambda_i A)^{k_i}.$$

- (iv) $\forall i$: Compute a basis $v_1^{(i)}, \dots, v_{l_i}^{(i)}$ of V_i .
- (v) Construct the $S \in \text{GL}(\mathbb{K}; n)$ with column vectors

$$v_j^{(i)}, \quad (\lambda_i I_n - A)v_j^{(i)}, \quad \dots, \quad (\lambda_i I_n - A)^{k_i-1}v_j^{(i)}.$$

Refresher III

— Inner Product Spaces —

- ▶ Inner Products
- ▶ Orthogonal Complements, Orthonormal Basis
- ▶ Gram–Schmidt Procedure
- ▶ QR decomposition, Least Squares Approximation
- ▶ Isometries, Self-Adjoint maps
- ▶ Spectral theorem
- ▶ Singular Value Decomposition

Inner Product Spaces

Definition

An **inner product space** is a vector space V over $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$ endowed with an **inner product** $\langle \cdot, \cdot \rangle: V \times V \rightarrow \mathbb{K}$ that satisfies $\forall u, v, w \in V$ and $\lambda \in \mathbb{K}$:

- ▶ (symmetry) $\langle v, w \rangle = \overline{\langle w, v \rangle}$
- ▶ (linearity) $\langle v + \lambda w, u \rangle = \langle v, u \rangle + \lambda \langle w, u \rangle$
- ▶ (definiteness) $\langle v, v \rangle > 0$ if $v \neq 0$.

If $\mathbb{K} = \mathbb{R}$, V is also called an **Euclidean space**, and if $\mathbb{K} = \mathbb{C}$, V is an **unitary space**.

Note: For the rest of *Refresher III* we always assume $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$!

Inner Product Spaces: Examples

Example

The **standard inner product** on $\mathbb{K}^n \simeq \mathbb{K}^{n,1}$ is $\langle v, w \rangle = \bar{w}^T v = \sum_{i=1}^n \bar{v}_i w_i$.

Example

For $M \in \mathbb{K}^{n,n}$ let $M^\dagger := M^* := \bar{M}^T$. Then $\langle M, N \rangle = \text{tr}(N^H M)$ is an inner product on $\mathbb{K}^{n,n}$.

Example

The L^2 product on the space of smooth functions $[0, 1] \rightarrow \mathbb{C}$ is

$$\langle f, g \rangle = \int_0^1 f(t) \overline{g(t)} dt.$$

Orthogonality: Definitions

Definition

- ▶ $v, w \in V$ are **orthogonal** if $\langle v, w \rangle = 0$.
- ▶ A basis $\mathcal{B} \subset V$ is **orthogonal** if all pairs $v, w \in \mathcal{B}$, $v \neq w$, are orthogonal.
- ▶ This basis is **orthonormal** if moreover $1 = \|v\| := \sqrt{\langle v, v \rangle}$ for all $v \in \mathcal{B}$.
- ▶ The **orthogonal complement** of a subspace $U \subset V$ is

$$U^\perp := \{v \in V \mid \langle u, v \rangle = 0 \ \forall u \in U\}.$$

Orthogonality: Basic Properties

Proposition

- ▶ The orthogonal complement $U^\perp$ is always a linear subspace, and $U \cap U^\perp = \{0\}$.
- ▶ All pairs $v, w \in V$ satisfy the [Cauchy-Schwarz inequality](#)

$$|\langle v, w \rangle| \leq \|v\| \|w\|.$$

- ▶ If $v, w \in V$ are orthogonal, then the [Pythagorean theorem](#) holds:

$$\|v + w\|^2 = \|v\|^2 + \|w\|^2.$$

Grahm–Schmidt Procedure

Theorem (Grahm–Schmidt)

Let $(V, \langle \cdot, \cdot \rangle)$ be a finite dimensional inner product space with basis $\{u_1, \dots, u_n\}$. We can inductively build an orthonormal basis $\{q_1, \dots, q_n\}$:

$$q_1 := u_1 / \|u_1\|,$$
$$q'_i := u_i - \sum_{k=1}^i \langle u_i, q_k \rangle q_k, \quad q_i := q'_i / \|q'_i\|.$$

Orthogonal Projection

Proposition

Let $(V, \langle \cdot, \cdot \rangle)$ a finite dimensional inner product space. Every linear subspace $U \neq \{0\}$ possesses an orthonormal basis $\{q_1, \dots, q_m\}$. The orthogonal projection onto U is given by

$$P_U(v) := \sum_{i=1}^m \langle v, q_i \rangle q_i \quad \forall v \in V.$$

It satisfies $P_U^2 = P_U$, $\text{im}(P_U) = U$, and $\ker(P_U) = U^\perp$.

Orthogonal Projection: Examples

Exercise

Let $\lambda_1, \dots, \lambda_m \in \mathbb{R}_+^m$ and $D \in \mathbb{R}^{m,m}$ be the diagonal matrix with $D_{ii} = \lambda_i$. Prove that $\langle v, w \rangle_D := w^* D v$ is an inner product on $\mathbb{K}^m$.

Proposition

Let $U \neq \{0\}$ be a subspace of $\mathbb{K}^m$ with $\langle \cdot, \cdot \rangle_D$ -orthonormal basis $\{u_1, \dots, u_n\}$. Then the matrix representation of P_U w.r.t. the standard basis of $\mathbb{K}^n$ is

$$[P_U] = \underbrace{\begin{pmatrix} | & & | \\ u_1 & \dots & u_n \\ | & & | \end{pmatrix}}_{:=A} \begin{pmatrix} - & u_1^* & - \\ & \vdots & \\ - & u_n^* & - \end{pmatrix} D.$$

If $\{u_1, \dots, u_n\}$ is a general basis, this becomes $[P_U] = A(A^* D A)^{-1} A^* D$.

Orthogonal Projection: Corollaries

Corollary

Let $(V, \langle \cdot, \cdot \rangle)$ be a finite dimensional inner product space and $U \subset V$ a linear subspace. Then $V = U \oplus U^\perp$.

Proposition (Distance Minimization)

Let $(V, \langle \cdot, \cdot \rangle)$ a finite dimensional inner product space and $U \neq \{0\}$ a subspace. For every $v \in V$ holds

$$\operatorname{argmin}_{u \in U} \|v - u\| = P_U(v).$$

Orthogonal & Unitary Matrices

Definition

- ▶ $Q \in \mathbb{R}^{n,n}$ is called **orthogonal** if $Q^T Q = I_n$. The set of orthogonal matrices is $O(n)$ and the subset of matrices with $\det(Q) = 1$ is $SO(n)$.
- ▶ $U \in \mathbb{C}^{n,n}$ is called **unitary** if $U^* U = I_n$. The set of unitary matrices is $U(n)$ and the subset of matrices with $\det(U) = 1$ is $SU(n)$.

Proposition

- ▶ Every orthogonal Q is unitary.
- ▶ A matrix $\mathbb{K}^{n,n}$ is unitary *if and only if* its columns are orthonormal w.r.t. the standard inner product on $\mathbb{K}^n$.
- ▶ $O(n)$, $SO(n)$, $U(n)$, and $SU(n)$ are subgroups of $GL(n)$.

QR Decomposition

Theorem (QR Decomposition)

Let $A \in \mathbb{K}^{m,n}$.

- ▶ There always exists a unitary matrix $Q \in U(m)$ and an upper triangular matrix $R \in \mathbb{K}^{m,n}$, i.e., $R_{ij} = 0$ if $i > j$, such that $A = QR$.
- ▶ If A has full rank, i.e., $\text{rank}(A) = \min\{m, n\}$, then this **QR decomposition** is unique.
- ▶ If A has full rank and $m \geq n$, then there is $Q \in \mathbb{K}^{m,n}$ and $R \in \text{GL}(\mathbb{K}; n)$ such that $A = QR$ and the columns of Q are an orthonormal basis for $\text{im}(Q)$.
This is called the **reduced QR decomposition**.

Least Squares Approximation

Proposition

Let $m \geq n$, $A \in \mathbb{R}^{m,n}$, and $b \in \mathbb{R}^m$. Denote by $Q \in \mathbb{R}^{m,n}$ and $R \in \text{GL}(\mathbb{K}; n)$ the reduced QR decomposition of A . Then the **least squares approximation** of the linear system “ $Ax = b$ ” is

$$\operatorname{argmin}_{x \in \mathbb{R}^n} \|Ax - b\| = R^{-1}Q^*b.$$

Note: Computing an inverse is usually **numerically unstable**. It should be avoided if possible. Luckily R is an upper triangle matrix. Thus, we can find $x \in \mathbb{R}^n$ with $Rx = Q^*b$ by **backward substitution**.

Polynomial Regression

Definition

Let $n, m \in \mathbb{N}$ and $x_0 < x_1 < \dots < x_m$. The **Vandermonde matrix** of these points is

$$\begin{pmatrix} 1 & x_0 & x_0^2 & \dots & x_0^n \\ 1 & x_1 & x_1^2 & \dots & x_1^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_m & x_m^2 & \dots & x_m^n \end{pmatrix} \in \mathbb{R}^{(m+1), (n+1)}$$

Proposition

The Vandermonde matrix has full rank.

Polynomial Regression

Theorem

Let $m \geq n$, $x_0 < x_1 < \dots < x_m$, and $y \in \mathbb{R}^{m+1}$. For $\alpha \in \mathbb{R}^{n+1}$ let $p_\alpha(x) = \sum_{i=0}^n \alpha_i x^i$. Let $Q \in \mathbb{R}^{m+1, n+1}$ and $R \in \text{GL}(\mathbb{R}; n+1)$ be the reduced QR decomposition of the Vandermonde matrix of the x_i . Then

$$\operatorname{argmin}_{\alpha \in \mathbb{R}^{n+1}} \sum_{i=0}^m |y_i - p_\alpha(x_i)|^2 = R^{-1} Q^T y.$$

Isometries

Definition

A linear map $f \in \text{Lin}(V, W)$ between inner product spaces $(V, \langle \cdot, \cdot \rangle_V)$ and $(W, \langle \cdot, \cdot \rangle_W)$ is an **isometry** if

$$\langle f(v_1), f(v_2) \rangle_W = \langle v_1, v_2 \rangle_V \quad \forall v_1, v_2 \in V.$$

Isometries: Properties

Proposition

Let $(V, \langle \cdot, \cdot \rangle_V)$ and $(W, \langle \cdot, \cdot \rangle_W)$ be inner product spaces, and $f \in \text{Lin}(V, W)$. The following are equivalent:

- ▶ f is an isometry.
- ▶ For all $v \in V$: $\|v\|_V = 1 \Rightarrow \|f(v)\|_W = 1$.
- ▶ For all $v \in V$: $\|v\|_V = \|f(v)\|_W$.
- ▶ If $\{b_1, \dots, b_n\}$ is orthonormal in V , then $\{f(b_1), \dots, f(b_n)\}$ in W orthonormal.

Isometries: Corollaries

Corollary

Every isometry $f \in \text{Lin}(V, W)$ is injective.

Corollary

Let $f \in \text{Lin}(V, V)$ be an isometry and $\mathcal{B} = \{b_1, \dots, b_n\}$ be an orthonormal basis of V . Then

- ▶ $\Phi_V: V \rightarrow \mathbb{K}^n$ given by $b_i \mapsto e_i$ is an isometry, where e_i are the standard basis vectors of $\mathbb{K}^n$ which we equip with the standard dot product.
- ▶ The matrix representation $[f]_{\mathcal{B}\mathcal{B}}$ of f is unitary, i.e., $[f]_{\mathcal{B}\mathcal{B}} \in \text{U}(n)$.
- ▶ f is invertible with $[f^{-1}]_{\mathcal{B}\mathcal{B}} = [f]_{\mathcal{B}\mathcal{B}}^{-1} = [f]_{\mathcal{B}\mathcal{B}}^*$.

The Adjoint Map

Definition

Let $f \in \text{Lin}(V, W)$ be a map between inner product spaces.

- ▶ A map $f^* \in \text{Lin}(W, V)$ is called **adjoint to f** if

$$\langle f(v), w \rangle_W = \langle v, f^*(w) \rangle_V$$

for all $v \in V$ and $w \in W$.

- ▶ If $W = V$ and $f = f^*$, then we call f **self-adjoint**.

The Adjoint Map: Properties

Proposition

Let $f \in \text{Lin}(V, W)$.

- ▶ f possesses at most one adjoint.
- ▶ If f^* is the adjoint of f , then f is the adjoint of f^* , i.e., $(f^*)^* = f$.
- ▶ If f is an isometry and $\dim(V) = \dim(W) < \infty$, then f^{-1} is the adjoint of f .

The Adjoint Map: Properties

Proposition

Let $f \in \text{Lin}(V, W)$.

- ▶ If $\{b_1, \dots, b_n\}$ is an orthonormal basis of V , then the adjoint of f is given by

$$f^*(w) := \sum_{i=1}^n \langle w, f(b_i) \rangle_W b_i \quad \forall w \in W.$$

- ▶ The matrix representation of the adjoint w.r.t. (finite) orthonormal bases $\mathcal{B}_V$ and $\mathcal{B}_W$ satisfies $[f^*]_{\mathcal{B}_W \mathcal{B}_V} = [f]_{\mathcal{B}_V \mathcal{B}_W}^*$.
- ▶ If $\dim(V) < \infty$, f is self-adjoint if and only if $[f]_{\mathcal{B}_V \mathcal{B}_V}$ is **hermitian** ($\mathbb{K} = \mathbb{C}$) or **symmetric** ($\mathbb{K} = \mathbb{R}$).

Spectral Theorem

Theorem (Spectral Theorem)

Let V be an inner product space over $\mathbb{C}$ with $\dim(V) < \infty$, and $f \in \text{Lin}(V, V)$. The following are equivalent

- ▶ $f \circ f^* = f^* \circ f$.
- ▶ *There exists an orthonormal basis of V that only consists of eigenvectors of f .*

Note: All self-adjoint maps and isometries satisfy the first condition!

Spectral Theorem: Corollaries

Proposition

- ▶ For every hermitian matrix $A \in \mathbb{C}^{n,n}$, i.e., $A = A^*$, there exists a diagonal matrix $D \in \mathbb{R}^{n,n}$ and a unitary matrix $U \in \mathcal{U}(n)$ such that $A = UDU^*$.
- ▶ For every symmetric matrix $A \in \mathbb{R}^{n,n}$, i.e., $A = A^T$, there exists a diagonal matrix $D \in \mathbb{R}^{n,n}$ and an orthogonal matrix $Q \in \mathcal{O}(n)$ such that $A = QDQ^T$.